

Introduction to Lagrange Multipliers

Toy example and generalization

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Problem statement

We want to maximise a function $f(x_1, \dots, x_D)$ subject to an equality constraint

$$g(x_1, \dots, x_D) = c.$$

In vector notation, with $\mathbf{x} = (x_1, \dots, x_D)^\top$:

$$\max_{\mathbf{x} \in \mathbb{R}^D} f(\mathbf{x}) \quad \text{subject to} \quad g(\mathbf{x}) = c.$$

1 A two-dimensional example

Let us take $D = 2$ and consider

$$f(x_1, x_2) = 2x_1 + x_2 \quad \text{subject to} \quad x_1^2 + x_2^2 = 1 \quad (\text{i.e. } \|\mathbf{x}\|_2^2 = 1).$$

The *feasible set* is the unit circle, and we are looking for the point of the circle where f is largest.

1.1 Geometric intuition: contour lines

The contour line (level set) of f at level k is the set of points where $f(x_1, x_2) = k$, i.e. the straight line

$$2x_1 + x_2 = k \iff x_2 = -2x_1 + k.$$

- $f(\mathbf{x}) = 0$: the line $x_2 = -2x_1$ passes through the origin. It crosses the circle at two points, which therefore satisfy the constraint, but $f = 0$ is clearly not the maximum (see Figure 1).
- $f(\mathbf{x}) = 2$: the line $x_2 = -2x_1 + 2$ still crosses the circle at two points, so the value 2 is reachable – and obviously "better" than $f = 0$.
- $f(\mathbf{x}) = 3$: the line $x_2 = -2x_1 + 3$ no longer meets the circle; it is *not possible* to satisfy the constraint with $f = 3$.

Hence the maximum reachable value of f on the circle lies between 2 and 3.

As k increases, the contour line slides in the direction of ∇f , the gradient of f . The largest value of k for which the line still touches the circle is the one where the line is *tangent* to the circle (Figure 1).

1.2 The tangency condition

Idea. If $f(\mathbf{x})$ is maximised while satisfying $g(\mathbf{x}) = c$, then the contour line of f through the optimum is tangent to the contour line $g(\mathbf{x}) = c$.

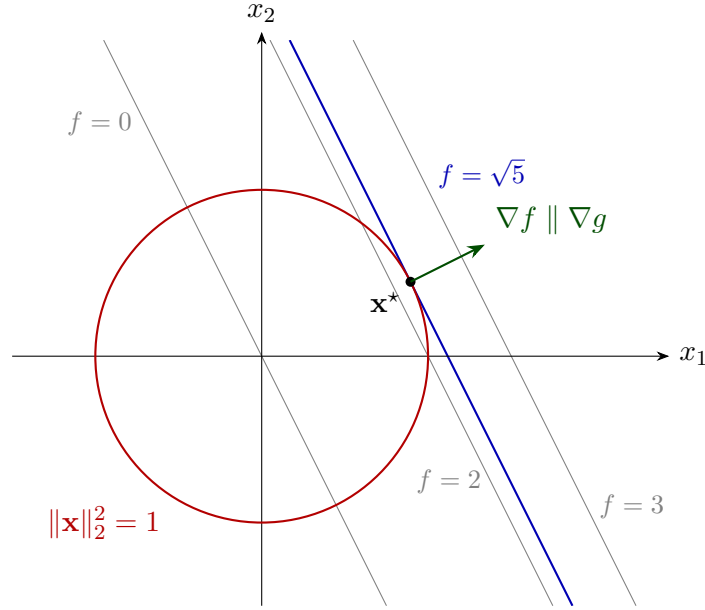


Figure 1: Contour lines of $f(x_1, x_2) = 2x_1 + x_2$ and the constraint $x_1^2 + x_2^2 = 1$. At the optimum $\mathbf{x}^*$, the contour line of f is tangent to the circle and the two gradients are parallel.

Gradients. The gradient $\nabla_{\mathbf{x}} f = \frac{\partial f}{\partial \mathbf{x}}$ at a point (x_1, x_2) is perpendicular to the contour line of f through that point (and likewise for g). If the two contour lines are tangent, they share the same tangent direction, so their normals—the gradients—are parallel:

$$\frac{\partial f}{\partial \mathbf{x}}(x_1, x_2) = \lambda \frac{\partial g}{\partial \mathbf{x}}(x_1, x_2) \quad \text{i.e.} \quad \nabla_{\mathbf{x}} f = \lambda \nabla_{\mathbf{x}} g$$

for some scalar $\lambda \in \mathbb{R}$, called the *Lagrange multiplier*. We call this equation the *tangency condition*.

1.3 Solving the example

We write the constraint as $g(x_1, x_2) = x_1^2 + x_2^2 - 1 = 0$ (i.e. $c = 0$). The gradients are

$$\begin{aligned} \frac{\partial f}{\partial \mathbf{x}}(x_1, x_2) &= \begin{pmatrix} \frac{\partial}{\partial x_1}(2x_1 + x_2) \\ \frac{\partial}{\partial x_2}(2x_1 + x_2) \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \\ \frac{\partial g}{\partial \mathbf{x}}(x_1, x_2) &= \begin{pmatrix} \frac{\partial}{\partial x_1}(x_1^2 + x_2^2 - 1) \\ \frac{\partial}{\partial x_2}(x_1^2 + x_2^2 - 1) \end{pmatrix} = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix} = 2\mathbf{x}. \end{aligned}$$

The tangency condition therefore reads

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \lambda \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}.$$

Together with the constraint, we obtain a system of 3 equations in 3 unknowns (x_1, x_2, λ) :

$$\begin{cases} x_1^2 + x_2^2 - 1 = 0 \\ 2\lambda x_1 - 2 = 0 \\ 2\lambda x_2 - 1 = 0 \end{cases}$$

From the last two equations, $x_1 = 1/\lambda$ and $x_2 = 1/(2\lambda)$. Substituting into the constraint gives

$$\frac{1}{\lambda^2} + \frac{1}{4\lambda^2} = 1 \iff \lambda^2 = \frac{5}{4} \iff \lambda = \pm \frac{\sqrt{5}}{2}.$$

Because of the square, there are two solutions:

$$\begin{cases} \lambda = \frac{\sqrt{5}}{2} \\ x_1 = \frac{2}{\sqrt{5}} \\ x_2 = \frac{1}{\sqrt{5}} \end{cases} \quad \text{and} \quad \begin{cases} \lambda = -\frac{\sqrt{5}}{2} \\ x_1 = -\frac{2}{\sqrt{5}} \\ x_2 = -\frac{1}{\sqrt{5}} \end{cases}$$

Evaluating f at both points:

$$f\left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right) = \frac{5}{\sqrt{5}} = \sqrt{5} \approx 2.236, \quad f\left(-\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right) = -\sqrt{5}.$$

The first point is the maximum and the second the minimum. Hence

$$f_{\max} = \sqrt{5} \approx 2.236,$$

a bit more than 2, as predicted by the geometric argument. This solves the example.

2 General case and the Lagrangian

Generic idea. The tangency condition $\nabla_{\mathbf{x}}f = \lambda\nabla_{\mathbf{x}}g$ remains valid when $D > 2$, and for arbitrary (differentiable) f and g —provided, of course, that a solution exists and that $\nabla_{\mathbf{x}}g \neq \mathbf{0}$ at the optimum.

Solving the constrained optimisation problem therefore comes down to solving a system of $D + 1$ equations in $D + 1$ unknowns $x_1, \dots, x_D, \lambda$:

$$\begin{cases} g(x_1, \dots, x_D) = c & (\text{constraint}) \\ \frac{\partial f}{\partial x_1}(x_1, \dots, x_D) = \lambda \frac{\partial g}{\partial x_1}(x_1, \dots, x_D) \\ \vdots & (\text{tangency condition}) \\ \frac{\partial f}{\partial x_D}(x_1, \dots, x_D) = \lambda \frac{\partial g}{\partial x_D}(x_1, \dots, x_D) \end{cases} \iff \begin{cases} g(\mathbf{x}) = c \\ \frac{\partial f}{\partial \mathbf{x}}(\mathbf{x}) = \lambda \frac{\partial g}{\partial \mathbf{x}}(\mathbf{x}) \end{cases}$$

Introducing the Lagrangian. Define the function of $D + 1$ inputs

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda(g(\mathbf{x}) - c).$$

Solving the system of $D + 1$ equations above is equivalent to solving

$$\nabla_{\mathbf{x}, \lambda} \mathcal{L}(\mathbf{x}, \lambda) = \mathbf{0}.$$

Indeed:

- $\frac{\partial \mathcal{L}}{\partial \lambda}(\mathbf{x}, \lambda) = -(g(\mathbf{x}) - c)$, so $\frac{\partial \mathcal{L}}{\partial \lambda} = 0 \iff g(\mathbf{x}) = c$ (the constraint);
- $\frac{\partial \mathcal{L}}{\partial \mathbf{x}}(\mathbf{x}, \lambda) = \frac{\partial f}{\partial \mathbf{x}}(\mathbf{x}) - \lambda \frac{\partial g}{\partial \mathbf{x}}(\mathbf{x})$, so $\frac{\partial \mathcal{L}}{\partial \mathbf{x}} = \mathbf{0} \iff \nabla_{\mathbf{x}}f = \lambda \nabla_{\mathbf{x}}g$ (the tangency condition).

3 Summary: steps to solve a constrained optimisation problem

To solve

$$\max_{\mathbf{x}} f(\mathbf{x}) \quad \text{subject to} \quad g(\mathbf{x}) = c :$$

1. Introduce the Lagrangian $\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda(g(\mathbf{x}) - c)$.
2. Find the critical points $(\mathbf{x}, \lambda)$ such that $\nabla_{\mathbf{x}, \lambda} \mathcal{L}(\mathbf{x}, \lambda) = \mathbf{0}$ (a system of up to $D+1$ equations).
3. Plug each critical point $\mathbf{x}$ into f and keep the one giving the maximum.